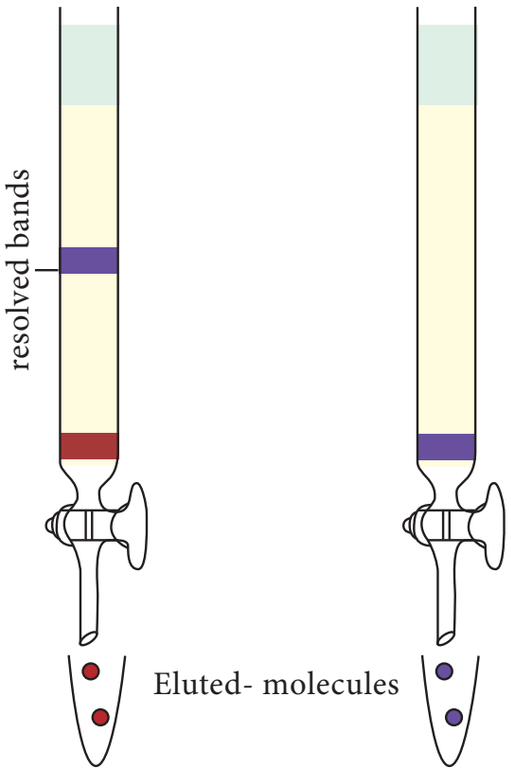


Fig.10.3 Column chromatography

10.3.2 Paper chromatography

Paper chromatography was discovered by the Russian investigators Izmailov and Schraiber. The filter paper strips, made up of cellulose are used to support the stationary phase (water) while the mobile organic phase moves on the suspended paper strip in a chamber. In this technique, the solution containing the mixture of compounds is spotted at a point above 3cm from the end of a Whatmann filter paper (Fig.10.4). After spotting the sample, the paper is dried and equilibrated into a glass jar which contains the desired solvent, for example butanol: acetic acid: water in the ratio of 4:1:5, which acts as a mobile phase. It should be ensured that the point of application remains well above the level of the solvent in the jar. The paper is suspended in such a manner so that it hangs freely without touching the sides of the container. The solvent ascends on the paper and is called as ascending chromatography. The factor governing the separation of compounds in the sample depends on the partition between two immiscible phases.

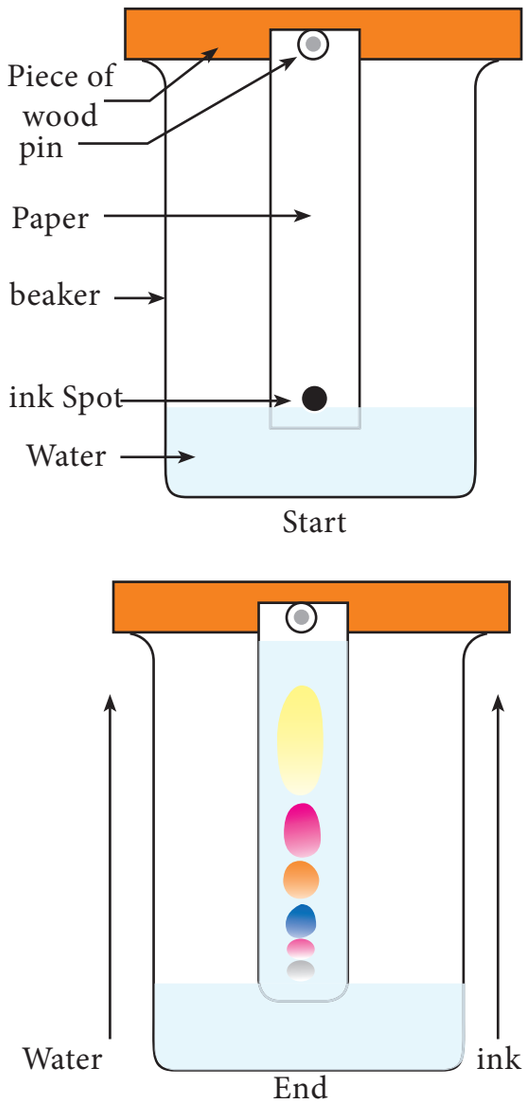


Fig.10.4 : Paper chromatography

When the solvent reaches the top of the paper, the paper is removed and the distance to which the solvent travelled is marked. In order to locate the compounds, the paper strip is dried and sprayed with

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